

Georg von Westphalen

INTENSIVE CARE MEDICINE: THE VITAL VIGIL

Titrating the threshold between life and the void. Stay fluffy!



TAPE-01

ARDS-the-Alveolar-Axe

Acute Respiratory Distress Syndrome

STAT: P/F Ratio < 300
Bilateral Opacities
Not heart failure



POWER: The Wet-Lung-Wallop. Axe floods the alveoli with protein-rich Sludge, turning the lungs into stiff, unexpandable Stone.

LORE: Cytokine-driven damage to the alveolar-capillary barrier creates Baby Lungs—small patches of functional tissue in a sea of collapse.

TAPE-01

TAPE-03

Vent-the-Vandal

Ventilator Induced Lung Injury

STAT: Plateau Pressure > 30 cmH₂O
Driving Pressure > 15



POWER: The Over-Stretch. Pushes too much air into too little lung, tearing the tissue apart.

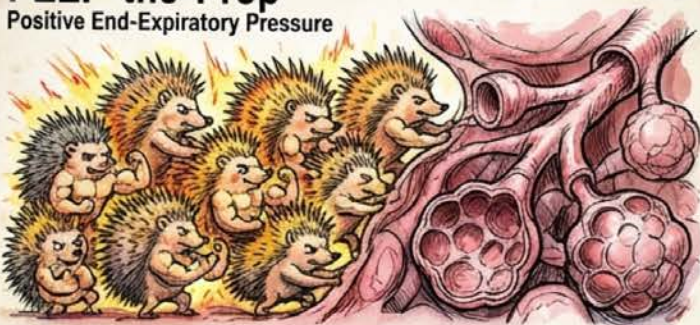
LORE: Volutrauma and Barotrauma. Turns Protective-Ventilation into a destructive Air-Hammer.

TAPE-03

PEEP-the-Prop

Positive End-Expiratory Pressure

STAT: 5-20 cmH₂O



POWER: The Balloon-Support. Keeps the tiny Alveolar-Balloons from snapping shut at the end of every breath.

LORE: Increases Functional Residual Capacity. Caveat: Too much PEEP crushes the Heart-Pump (Decreases Preload).



TAPE-02

Stay fluffy!

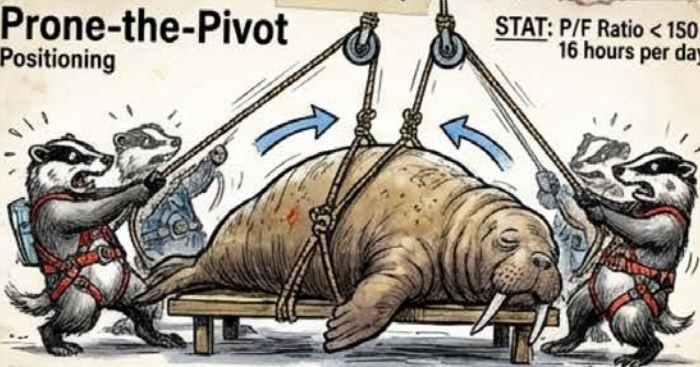


TAPE-04

Prone-the-Pivot

Positioning

STAT: P/F Ratio < 150
16 hours per day



POWER: The Gravity-Flip. Let the Back-Lungs breathe while the Front-Lungs rest.

LORE: Homogenizes stress and strain. Moves the weight of the heart and gut off the dorsal alveoli to improve the Ventilation-Perfusion-Match.

TAPE-04

The Sabotaged Sketchbook

Norad-the-North-Star

Norepinephrine

STAT: Alpha-1 Agonist
First-line for Sepsis



POWER: The Pressure-Anchor. Clamps the leaky Peripheral-Pipes to ensure the Brain-and-Heart-Vaults stay pressurized.

LORE: The Vasopressor of Choice. Increases MAP without over-revving the heart-engine.

Art-Line-the-Archer

Arterial Line

STAT: Continuous Waveform
Radial or Femoral



POWER: The Real-Time-Radar. Gives the Beat-by-Beat truth, spotting every Pulse-Pressure-Variation.

LORE: Allows for frequent blood gases. If the waveform Dampens, the Archer is hitting a wall or pipes are kinked.

Dobuta-the-Dynamo

Dobutamine

STAT: Beta-1 Agonist
Positive Inotrope



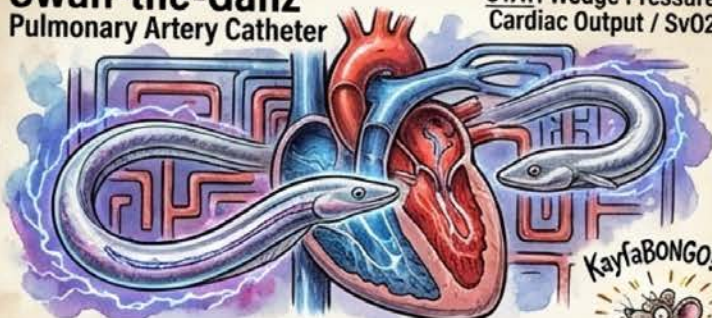
POWER: The Piston-Push. Screams at the failing heart-engine to squeeze harder.

LORE: Used in Cardiogenic-Carl. Caveat: It's a vasodilator—he might drop the MAP while boosting the flow.

Swan-the-Ganz

Pulmonary Artery Catheter

STAT: Wedge Pressure
Cardiac Output / SvO2



POWER: The Deep-Diver. Swims through the heart into the lungs to see the pressure in the Left-Atrium-Lobby.

LORE: Differentiates Heart-Failure-Flood from Sepsis-Leak. Rare now, but essential for Complex-Congestion.

CRRT-the-Constant

Continuous Renal Replacement Therapy

STAT: Ultrafiltration Rate
Dialysate Flow

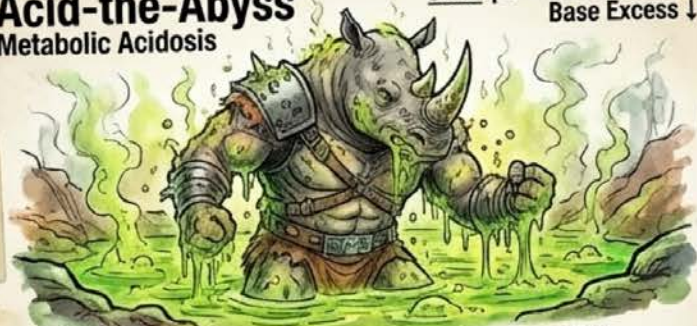


POWER: The 24-Hour-Sieve. Slowly cleans the blood, removing Toxin-Tokens without crashing the blood pressure.
LORE: Gentle. Mimics the natural kidney for wrestlers who are too unstable for a fast Flush.

Acid-the-Abyss

Metabolic Acidosis

STAT: pH < 7.2 / Lactate ↑
Base Excess ↓



POWER: The Protein-Melter. Turns the blood sour, causing Pressor-Pumps to stop and the heart to stumble.
LORE: A sign of Tissue-Starvation. If the cells can't breathe, they vomit acid. A signal of Energy-Failure.

Diagnosis?!



Hyper-K-the-Killer

Potassium

STAT: K+ > 6.5
Peaked T-waves
QRS widening



POWER: The Electrical-Eraser. Shorts out the heart's wiring, leading to the Sine-Wave-Silence.
LORE: Stops the resting membrane potential. Treat with Calcium-the-Shield to protect wires, then Insulin-the-Evictor.

Citrate-the-Clot-Stopper

Regional Anticoagulation

STAT: Regional Anti-coagulation for CRRT



POWER: The Machine-Shield. Eats the calcium in the dialysis circuit so blood doesn't Clot-the-Filter.
LORE: Regional means it only works inside the machine.
Caveat: If the liver can't process it, causes Calcium-Crashing.

The Sabotaged Sketchbook

Septic-Sam The Systemic Riot

STAT: qSOFA ≥ 2
Source of infection
MAP < 65



POWER: The Capillary-Collapse. Opens all Volume-Valves and turns vessel-walls into Leaky-Cheese.

ORE: Immune-overdrive causes massive vasodilation. Blood hides in Peripheral-Puddles, leaving the brain dry.

Lactate-the-Debt-Collector Hypoxia Marker

STAT: Value > 2 mmol/L
Clearance tracking



POWER: The Hypoxia-Tax. Represents the Oxygen-Debt from failing aerobic metabolism.

LORE: High Lactate = Tissue Ischemia. Clearing the Lactate proves the Resuscitation-Rescue is winning.

Procal-the-Predictor Procalcitonin

STAT: Elevated in
bacterial infection
Serial monitoring



POWER: The Bacterial-Beacon. Glows bright when Bacterial-Punks are in the blood, dim for Viral-Vandals.

LORE: Guides when to start—and when to stop—the Antibiotic-Artillery.

BLÖDSINN!

ANTIBIOTICS
= FASCISM!



Source-Control-the-Striker Infection Eradication

STAT: Abscess drainage
Dead tissue removal
Line exchange



POWER: The Root-Removal. Knows that Antibiotic-Ammo fails if the Nest isn't destroyed.

LORE: Ubi pus, ibi evacua. (If there is pus, drain it). If the line is dirty, pull it.

The Sabotaged Sketchbook

ICP-the-Intracranial-Pressure Brain Swelling

STAT: Normal < 15 mmHg
CPP = MAP - ICP



POWER: The Skull-Squeeze. Swelling brain crushes the Oxygen-Entry-Vessels.
LORE: The skull is a Closed-Box. If ICP reaches MAP, the Brain-Death-Bell rings.

Propofol-the-Milk The Sedative

STAT: GABA-A Agonist
Rapid onset and offset



POWER: The Mainframe-Mute. Puts the brain to sleep and lowers the Oxygen-Burn-Rate.
LORE: Allows the Daily-Wake-Up-Test. Caveat: PRIS (Infusion Syndrome) causes acidosis and heart failure.

MedKayfabe!



Osmotic-the-Osmotic Mannitol / Hypertonic Saline

STAT: 20% Mannitol
3% Saline



POWER: The Water-Sucker. Pulls water out of swollen brain-tissue and into the blood.
LORE: Creates an Osmotic-Gradient. Caveat: Mannitol causes Diuresis-Drain, 3% Saline keeps volume in pipes.

Dex-the-Precedex Dexmedetomidine

STAT: Alpha-2 Agonist
Non-respiratory depressant



POWER: The Conscious-Calm. Provides Analgo-Sedation—sleepy but can answer the Neuro-Check.
LORE: Doesn't stop the breathing-drive. Great for Ventilator-Weaning. Caveat: Can cause Brady-Bill (Slow heart).

The Sabotaged Sketchbook

TPN-the-Total-Parental Nutrition

STAT: Lipids, Amino Acids, Dextrose
Central Line only



POWER: The Vein-Feed. Bypasses the Gut-Gate to feed directly through the blood.

LORE: The Last Resort. TPN loves the Liver-Limp and Line-Infection-Iggyes. If the gut works, use the gut!

Ileus-the-Idle The Gut-Strike

STAT: Absent bowel sounds
No passage of gas



POWER: The Static-Snake. Intestines stop moving, causing Belly-Bloat and Vomitus-Vex.

LORE: Often caused by Opioid-Overload or Electrolyte-Emptying.

Stress-Ulcer-the-Shadow Gastric Erosion

STAT: Risk:
Ventilation > 48h
Coagulopathy



POWER: The Acid-Erosion. Eats the stomach lining during the Stress-Storm, causing a Hidden-Bleed.

LORE: Prevented by PPI-the-Lock or H2-the-Hinderer.

Liver-the-Limp Acute Liver Failure

STAT: INR > 1.5
Encephalopathy
No prior cirrhosis



POWER: The Detox-Failure. Stops cleaning the blood, letting Ammonia-the-Fog cloud the brain.

LORE: Watch for Cerebral-Edema-Swelling. It is a race for the Transplant-Train.

HUMBUG!



The Sabotaged Sketchbook

DIC-the-Disseminated Coagulopathy

STAT: PT/aPTT ↑/
Fibrinogen ↓ / Platelets ↓
D-Dimer ↑↑



POWER: The Clot-and-Bleed. Uses up all Clot-Tokens in tiny, useless fights, leaving nothing for Major-Leaks.
LORE: A Consumption-Coagulopathy. Treat the underlying Sam-Sepsis to stop the spiral.

FFP-the-Flow Fresh Frozen Plasma

STAT: Replaces all clotting factors



POWER: The Liquid-Gold. Restores the Coagulation-Cascade when the INR-Alarm is ringing.
LORE: Essential for Wart-Reversal and Massive-Hemorrhage.

Platelet-the-Plug Cellular Clotting

STAT: Transfusion threshold
< 10,000 (Stable) or
< 50,000 (Bleeding)



POWER: The Hole-Stoppage. The first line of defense against Capillary-Leaks.
LORE: In the ICU, Plugs get eaten by Sam-Sepsis and Spleen-the-Sponge.

KayfaBINGO!



TEG-the-Thromboelastography Viscoelastic Testing

STAT: R-time, K-time,
Alpha-angle, MA



POWER: The Real-Time-Clot-Movie. Tells exactly why blood isn't clotting: Factor-Failure, Platelet-Paucity, or Fibrin-Frenzy?
LORE: Guided Goal-Directed-Therapy. Don't guess—TEG it!

Hydrocort-the-Sepsis-Support Corticosteroids

STAT: 200mg per day
For vasopressor-refractory shock



POWER: The Vessel-Sensitizer. Helps the Norad-Pressor work better by making vessels sensitive to the signal.
LORE: Replaces Relative-Adrenal-Insufficiency.

Sugamma-the-Reverser Sugammadex

STAT: Encapsulating agent for
Rocuronium/Vecuronium.



POWER: The Muscle-Waks-Up. Grabs Paralyzer-Punks and drags them away from the Nerve-Junction.
LORE: Immediate reversal of neuromuscular blockade. No Wet-Side-Effects like Neostigmine.

KayfaBOGGLE!



Dantrolene-the-Inferno-Cooler Muscle Relaxant

STAT: RyR1 Antagonist



POWER: The Calcium-Cutter. Stops the Muscle-Meltdown in Malignant Hyperthermia.
LORE: Prevents the runaway release of calcium from the Sarcoplasmic-Silo.

Fentanyl-the-Fast-Acting Synthetic Opioid

STAT: Synthetic Opioid /
100x potency of Morphine.



POWER: The Surgical-Calm. The go-to for continuous Analgesia-Infusion in the ventilated wrestler.
LORE: Rapid on/off. Caveat: Causes Chest-Wall-Rigidity-Stuff if pushed too fast.

The Sabotaged Sketchbook

Ringer-the-Balanced Ringer's Lactate

STAT: Balanced crystalloid
Contains Electrolytes



POWER: The Physiological-Fill. The Gold Standard for filling the Septic-Tank.

LORE: Closer to blood than Normal-Saline-the-Salty, which can cause Chloride-Acid-Abyss.

Albumin-the-Colloid Volume Expander

STAT: 5% or 20% solution
High Oncotic Pressure



POWER: The Water-Magnst. Stays in the Pipe-Highway and pulls fluid out of the Tissue-Swamps (Edema).
LORE: Used when crystalloids fail or after large-volume Ascites-Draining.

Diuresis-the-Dry-Down Fluid Removal

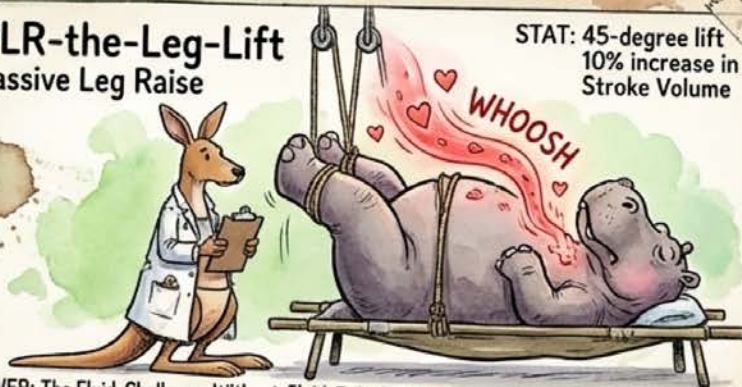
STAT: Negative Fluid
Balance goal



POWER: The Squeeze-Out. Removes the Extra-Weight post-shock to let the lungs breathe.
LORE: Lavitra-the-Lasix is the main tool. Being Too-Wet increases the Ventilator-Time.

PLR-the-Leg-Lift Passive Leg Raise

STAT: 45-degree lift
10% increase in
Stroke Volume



POWER: The Fluid-Challenge-Without-Fluid. Tells if the heart wants more volume before you pour it in.
LORE: The Reversible-Auto-Transfusion. If MAP goes up, the heart is Fluid-Responsive.

What the
FLUFF?!

Re-Intubate-the-Panic

Failed Extubation

STAT: Failed extubation / Stridor / High Work of Breathing



POWER: The Emergency-Return. When the Golden-Tube is removed too early and the Airway-Assassins return.
LORE: High-risk move. Swelling (Edema) makes the second pass much harder than the first.

Tension-the-Tony

ICU Redux

STAT: PEEP-induced / Sudden high airway pressure / Sudden drop in BP



POWER: The Machine-Induced-Crush. The ventilator Pops a lung-bleb, squashing the heart.
LORE: Always disconnect the vent first in sudden PEA to rule out Auto-PEEP or Tension-Tony.

HIT-the-Hunter

Heparin Induced Thrombocytopenia

STAT: 50% drop in platelets / 5-10 days after Heparin

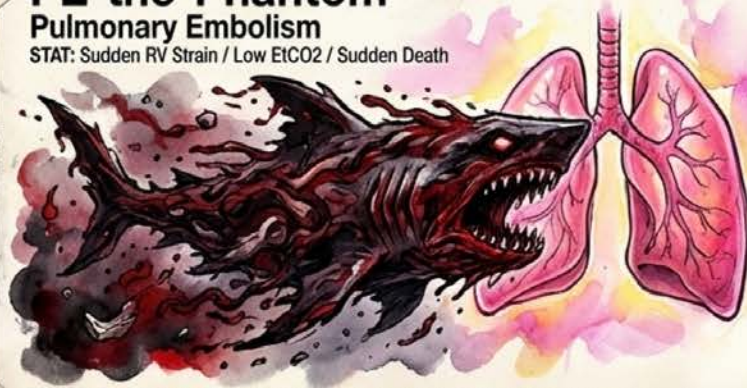


POWER: The Clot-Irony. Makes platelets clump together while numbers drop, causing Venous-and-Arterial-Jams.
LORE: Immune-mediated riot. Stop Heparin immediately and use Argatroban-the-Alternative.

PE-the-Phantom

Pulmonary Embolism

STAT: Sudden RV Strain / Low EtCO2 / Sudden Death



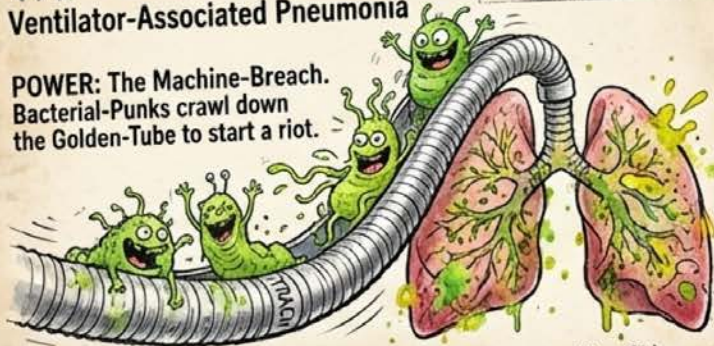
POWER: The Highway-Block. A clot-monster from the Leg-Deep-Veins jams the Lung-Entrance.
LORE: In the ICU, the Phantom is often completely silent until it is too late.

VAP-the-Vent-Pneumonia

Ventilator-Associated Pneumonia

STAT: New infiltrate
> 48h after intubation /
Fever / Secretions ↑

POWER: The Machine-Breach.
Bacterial-Punks crawl down
the Golden-Tube to start a riot.



LORE: Prevented by Head-Up-Position and Oral-Chlorhexidine-Chloe.

C-Diff-the-Colitis

Clostridioides difficile

STAT: Toxic Megacolon risk
/ Post-antibiotic diarrhea

POWER: The Gut-Explosion.
Takes over the Bowel-Basement
after Antibiotic-Artillery kills
the good guards.



LORE: Smells like a Horse-Barn. Treat with Vancomy-the-Vault (Oral).

CLABSI-the-Line-Infection

Central Line Associated Bloodstream Infection

STAT: Central Line
Associated Blood-
stream Infection

POWER: The IV-Invasion.
Staph-the-Skin-Steward
enters the Blood-Highway
through the CVC-Portal.



LORE: Scrub the Hub! If the line is infected, it must be Evicted.

ESKAPE-the-Superbugs

Multidrug Resistance

STAT: Enterococcus, Staph,
Klebsiella, Acinetobacter,
Pseudomonas, Enterobacter



POWER: The Armor-Plated-Punks. Multidrug-Resistance shields
laugh at standard Antibiotic-Ammo.

LORE: They use Efflux-Pumps and Enzyme-Blasters (Carbapenemases) to survive.

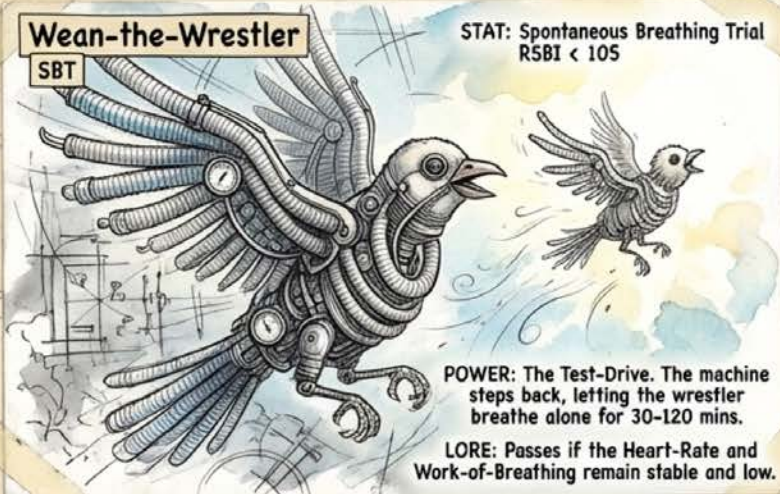
KoyfaBIZARRO!

KoyfaBIZARRO!

Wean-the-Wrestler

SBT

STAT: Spontaneous Breathing Trial
RSBI < 105



POWER: The Test-Drive. The machine steps back, letting the wrestler breathe alone for 30-120 mins.

LORE: Passes if the Heart-Rate and Work-of-Breathing remain stable and low.

PICS-the-Post-ICU-Struggle

Post-Intensive Care Syndrome

STAT: Post-Intensive Care Syndrome
Weakness + Cognitive drop



POWER: The Long-Shadow. The wrestler remains weak and forgetful for months after leaving the Vital-Vigil.

LORE: ICU-Acquired-Weakness. The muscles melted away during the long sleep.

Delirium-the-Daze

ICU Psychosis

STAT: CAM-ICU positive
Fluctuating consciousness



POWER: The Hallucination-Haze. Makes the wrestler see Shadow-Monsters and pull at their tubes.

LORE: Caused by Sleep-Sabotage, Benzodiazepine-Bouncers, and Sam-Sepsis.

Comfort-Care-the-Transition

Palliative Care

STAT: Palliative Extubation
Symptom control



POWER: The Peaceful-Exit. When the Scrimmage is lost, focus shifts to a Quiet-Mat without pain.

LORE: Withdrawal of Life-Sustaining Treatment. Morphine and Midazolam ensure the final breath is soft.

Stay fluffy!



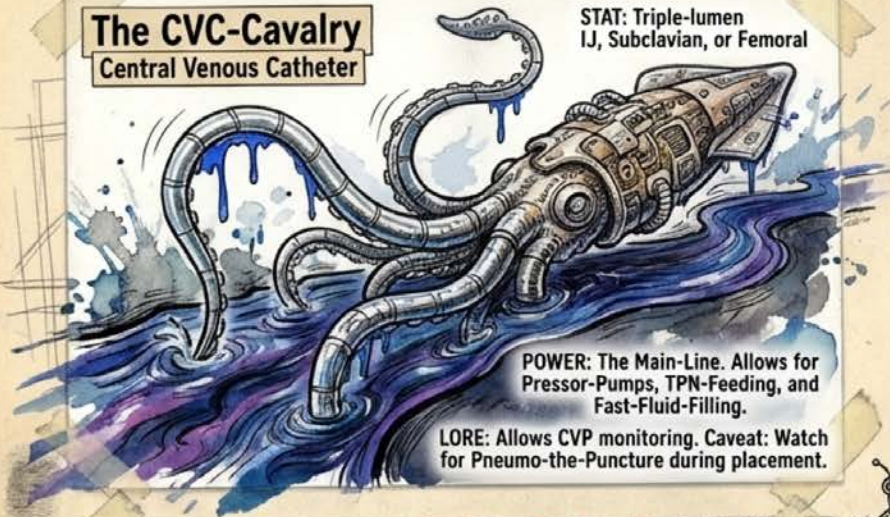
GW

Georg v. Westphalen

The CVC-Cavalry

Central Venous Catheter

STAT: Triple-lumen
IJ, Subclavian, or Femoral



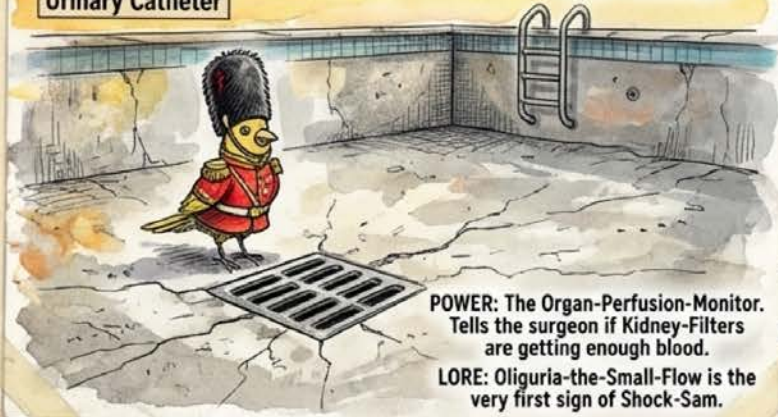
POWER: The Main-Line. Allows for
Pressor-Pumps, TPN-Feeding, and
Fast-Fluid-Filling.

LORE: Allows CVP monitoring. Caveat: Watch
for Pneumo-the-Puncture during placement.

The Foley-Sentry

Urinary Catheter

STAT: Hourly output tracking
Target > 0.5ml/kg/h



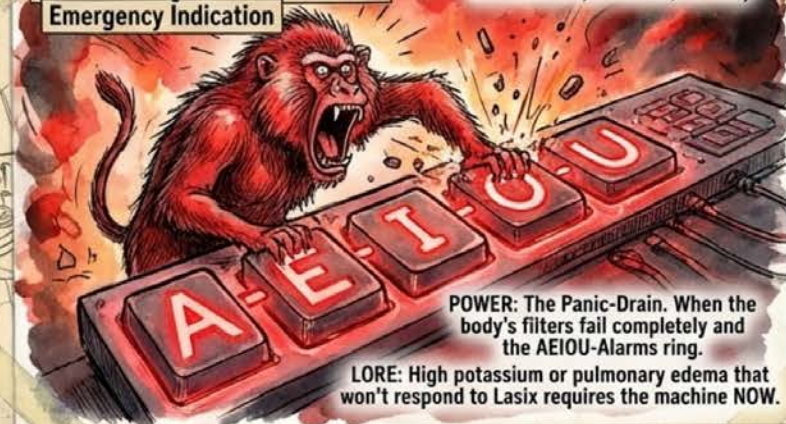
POWER: The Organ-Perfusion-Monitor.
Tells the surgeon if Kidney-Filters
are getting enough blood.

LORE: Oliguria-the-Small-Flow is the
very first sign of Shock-Sam.

The Dialysis-Disaster

Emergency Indication

STAT: AEIOU (Acidosis, Electrolytes,
Intoxications, Overload, Uremia)



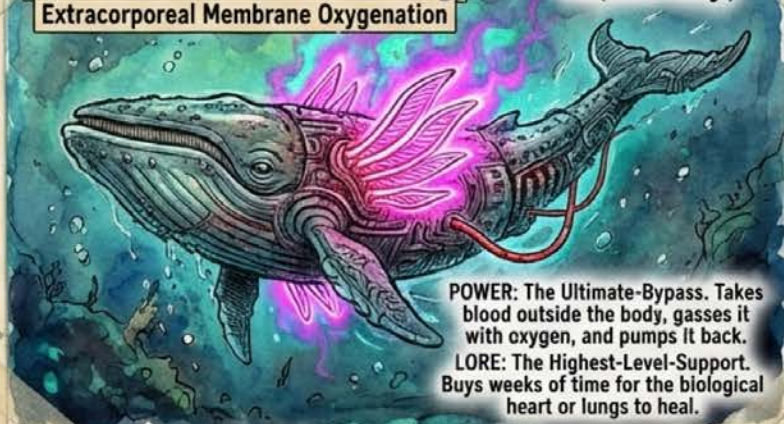
POWER: The Panic-Drain. When the
body's filters fail completely and
the AEIOU-Alarms ring.

LORE: High potassium or pulmonary edema that
won't respond to Lasix requires the machine NOW.

The ECMO-the-Extra-Lung

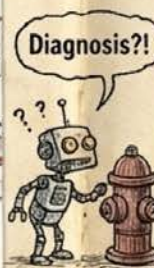
Extracorporeal Membrane Oxygenation

STAT: V-V (Lungs) vs.
V-A (Heart + Lungs)



POWER: The Ultimate-Bypass. Takes
blood outside the body, gasses it
with oxygen, and pumps it back.

LORE: The Highest-Level-Support.
Buys weeks of time for the biological
heart or lungs to heal.



The ICU Attending

Doc FrizzleBob

STAT: Intensivist /
Master of Multitasking.



POWER: The Meta-Physiologist.
LORE: Is the CO2 rising? Is the MAP holding? Is the Lactate clearing? Balance the system! Stay fluffy!

FrizzleBob's Quest 2

The Sudden-Silent-Code

STAT: Vent + Pressure Spike
+ HR 0 (PEA) + SpO2 drop

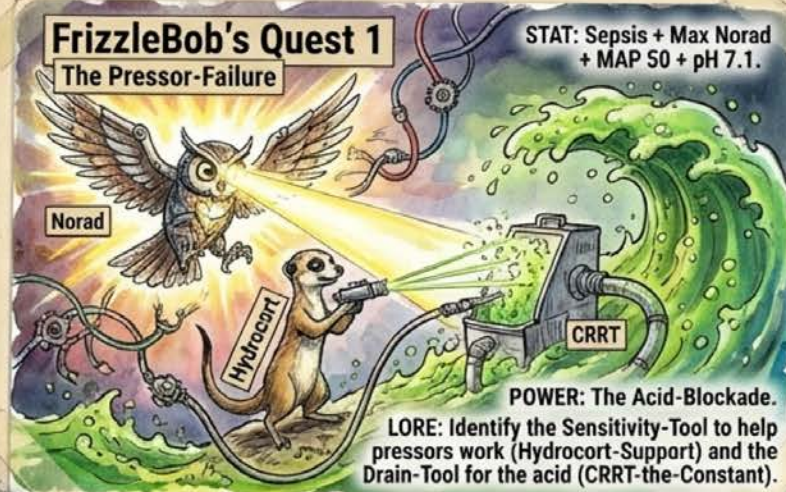


POWER: The Air-Trap.
LORE: Identify the Machine-Induced-Killer (Tension-the-Tony). The very first move: Disconnect the Vent!

FrizzleBob's Quest 1

The Pressor-Failure

STAT: Sepsis + Max Norad
+ MAP 50 + pH 7.1.



POWER: The Acid-Blockade.
LORE: Identify the Sensitivity-Tool to help pressors work (Hydrocort-Support) and the Drain-Tool for the acid (CRRT-the-Constant).

FrizzleBob's Quest 3

The Muscle-Melter

STAT: OR + EtCO2 off charts
+ Temp 41°C + Rigid muscles



POWER: The Sarcoplasmic-Fire.
LORE: Identify the Inland-Inferno (Malignant Hyperthermia) and deploy the Calcium-Cutter (Dantrolene-the-Inferno-Cooler).

Stay fluffy!
MedKayfabe!

Stay fluffy!
MedKayfabe!

